PROBLEM8:

PROBLEM STATEMENT:

Fuel Consumption and Route Comparison: Calculate fuel requirements and journey costs for several vehicles using distance, mileage, fuel prices, and reserve requirements; identify the lowest cost feasible option.

OBJECTIVE:

To develop a C program that calculates fuel requirements and journey costs for multiple vehicles using distance, mileage, fuel prices, and reserve fuel requirements. The program compares the total costs, checks which vehicles meet the fuel reserve requirement, and identifies the lowest-cost feasible vehicle.

ALGORITHM:

1. Start the program.
2. Input the number of vehicles to compare.
3. For each vehicle, input its name, mileage (km/litre), fuel price (₹/litre), and available fuel (litres).
4. Input the journey distance and required reserve fuel.
5. Validate all inputs to ensure they are positive and valid.
6. Calculate the fuel required for the journey:

Fuel required = Distance ÷ Mileage

7. Calculate the total fuel needed, including the reserve:

Total fuel needed = Fuel required + Reserve fuel

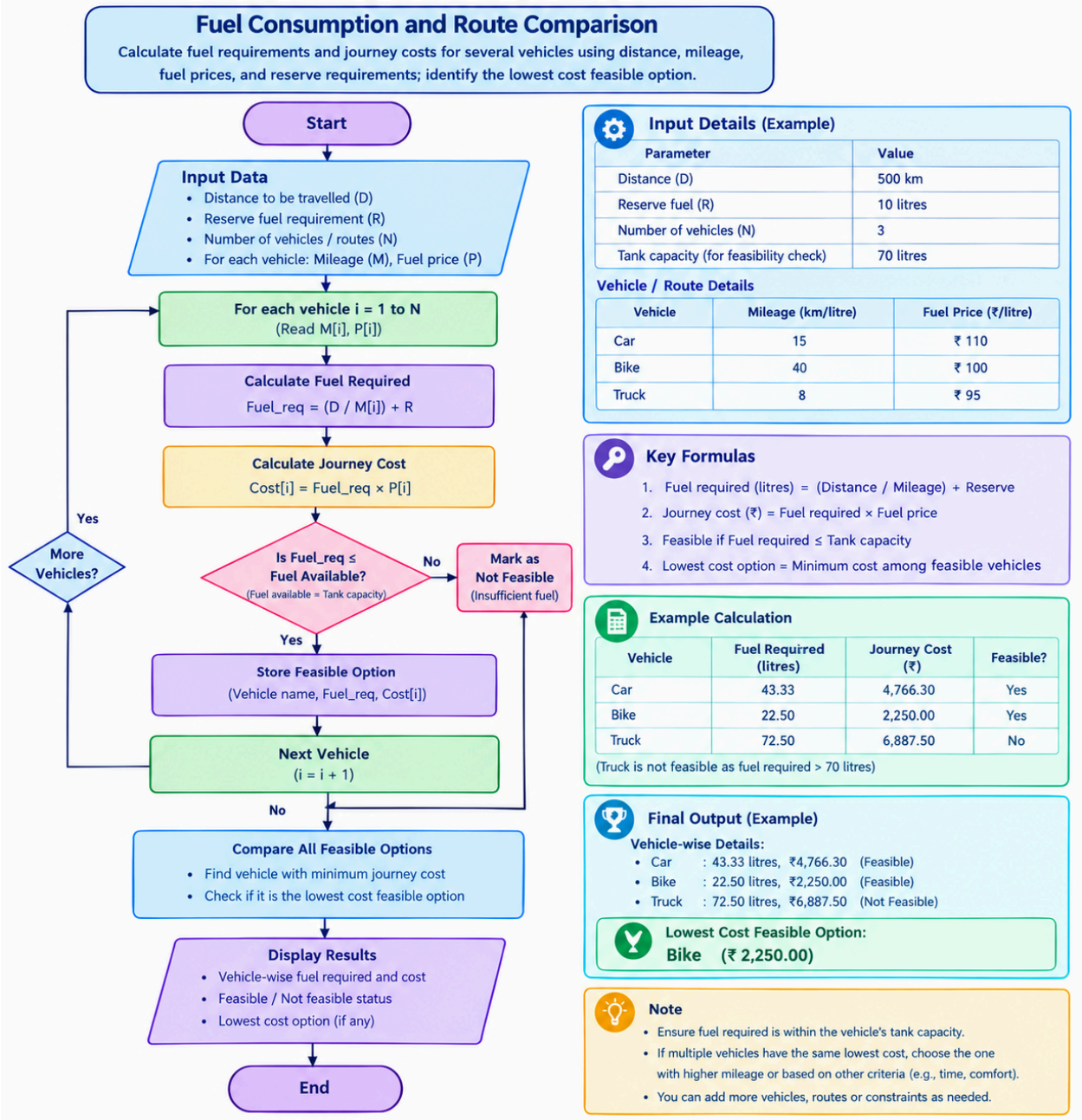
8. Calculate the journey cost:

Journey cost = Fuel required × Fuel price

9. Check whether the available fuel is sufficient to complete the journey and maintain the required reserve.
10. Compare the costs of all feasible vehicles.
11. Identify the feasible vehicle with the lowest journey cost.
12. Display the fuel requirements, journey costs, feasibility status, and lowest-cost option.

13. Stop.

FLOWCHART:



PROGRAM:

```c
#include <stdio.h>

int main() {
    int n, i, best = -1;
    char name[20][30];
    float distance, reserve;
    float mileage[20], price[20], available[20];
    float fuel[20], cost[20], totalFuel[20];
    int feasible[20];
    float lowestCost = 0;

    printf("Enter journey distance (km): ");
    scanf("%f", &distance);

    printf("Enter required reserve fuel (litres): ");
    scanf("%f", &reserve);

    printf("Enter number of vehicles (1-20): ");
    scanf("%d", &n);

    if (distance <= 0 || reserve < 0 ||
        n < 1 || n > 20) {
        printf("Invalid input!\n");
        return 0;
    }

    for (i = 0; i < n; i++) {
        printf("\nEnter vehicle name: ");
        scanf("%29s", name[i]);

        printf("Enter mileage (km/litre): ");
        scanf("%f", &mileage[i]);

        printf("Enter fuel price (Rs./litre): ");
        scanf("%f", &price[i]);

        printf("Enter available fuel (litres): ");
        scanf("%f", &available[i]);

        if (mileage[i] <= 0 || price[i] <= 0 ||
            available[i] < 0) {
            printf("Invalid vehicle details!\n");
            return 0;
        }
```

```
        fuel[i] = distance / mileage[i];
        totalFuel[i] = fuel[i] + reserve;
        cost[i] = fuel[i] * price[i];

        feasible[i] = (available[i] >= totalFuel[i]);

        if (feasible[i] &&
            (best == -1 || cost[i] < lowestCost)) {
            best = i;
            lowestCost = cost[i];
        }
    }

    printf("\n--- FUEL AND ROUTE COMPARISON ---\n");

    for (i = 0; i < n; i++) {
        printf("\nVehicle: %s\n", name[i]);
        printf("Fuel for journey: %.2f litres\n", fuel[i]);
        printf("Fuel including reserve: %.2f litres\n",
               totalFuel[i]);
        printf("Journey cost: Rs.%.2f\n", cost[i]);

        if (feasible[i])
            printf("Status: Feasible\n");
        else
            printf("Status: Not feasible\n");
    }

    printf("\n--- BEST OPTION ---\n");

    if (best == -1) {
        printf("No feasible vehicle available.\n");
    } else {
        printf("Lowest-cost vehicle: %s\n", name[best]);
        printf("Journey cost: Rs.%.2f\n", cost[best]);
        printf("Fuel required: %.2f litres\n", fuel[best]);
        printf("Reserve fuel: %.2f litres\n", reserve);
    }

    return 0;
}
```

PSEUDOCODE:

```
BEGIN

INPUT journey_distance
INPUT reserve_fuel
INPUT number_of_vehicles

IF distance <= 0 OR reserve_fuel < 0
  OR number_of_vehicles is invalid
   PRINT "Invalid input"
   STOP
END IF

FOR each vehicle
   INPUT vehicle_name
   INPUT mileage
   INPUT fuel_price
   INPUT available_fuel

   IF mileage <= 0 OR fuel_price <= 0
     OR available_fuel < 0
      PRINT "Invalid vehicle details"
      STOP
   END IF

   fuel_required = journey_distance / mileage

   total_fuel_needed = fuel_required + reserve_fuel

   journey_cost = fuel_required * fuel_price

   IF available_fuel >= total_fuel_needed
      feasible = TRUE
   ELSE
      feasible = FALSE
   END IF
END FOR

DISPLAY fuel requirements and journey costs

lowest_cost = INFINITY
best_vehicle = NONE
```

```
FOR each vehicle
   IF vehicle is feasible AND journey_cost < lowest_cost
      lowest_cost = journey_cost
      best_vehicle = vehicle
   END IF
END FOR

IF best_vehicle exists
   DISPLAY best_vehicle
   DISPLAY lowest_cost
ELSE
   DISPLAY "No feasible vehicle available"
END IF

STOP
```

OUTPUT:

Example: Journey distance = 300 km; reserve fuel = 5 litres

```
Enter journey distance (km): 300
Enter required reserve fuel (litres): 5
Enter number of vehicles (1-20): 3

Enter vehicle name: Car
Enter mileage (km/litre): 15
Enter fuel price (Rs./litre): 105
Enter available fuel (litres): 30

Enter vehicle name: Bike
Enter mileage (km/litre): 40
Enter fuel price (Rs./litre): 105
Enter available fuel (litres): 10

Enter vehicle name: SUV
Enter mileage (km/litre): 10
Enter fuel price (Rs./litre): 100
Enter available fuel (litres): 40

--- FUEL AND ROUTE COMPARISON ---

Vehicle: Car
Fuel for journey: 20.00 litres
```

Fuel including reserve: 25.00 litres
Journey cost: Rs.2100.00
Status: Feasible

Vehicle: Bike
Fuel for journey: 7.50 litres
Fuel including reserve: 12.50 litres
Journey cost: Rs.787.50
Status: Not feasible

Vehicle: SUV
Fuel for journey: 30.00 litres
Fuel including reserve: 35.00 litres
Journey cost: Rs.3000.00
Status: Feasible

--- BEST OPTION ---
Lowest-cost vehicle: Car
Journey cost: Rs.2100.00
Fuel required: 20.00 litres
Reserve fuel: 5.00 litres